

use characters that can host abbreviations with filenames. Using the pathname expansion, it becomes easy to specify multiple files all at once. This feature of pathname expansion is also known as 'globbing'. This is highly useful in the shell scripts of programs where you might not have precise ideas about the filename's existence. For example, you may want to process various log files named log0001, log0002 etc and despite the differing numbers, pathway expansion will match them all.

When any pattern is used for this expansion, then the character '.' at the beginning of the filename immediately following the slash has to match explicitly. Then the shell option is set to dot log. The file names ('.', '..') have to match explicitly even if the shell option is set. While making the file name the slash character is matched explicitly by the pattern, and in the other context, they can be matched with any special character. Filename expansions are not similar to the regular expressions which are mostly used by the utilities to specify the group of strings. Filename expansion can be concluded as a shorthand for specifying all the filenames having matching patterns at the base.

Tilde expansion – The Tilde character ('~') has a significant meaning. When it is used before the word then it expands the home directory of the user, and if the user is not named, then it expands the home directory of the current user. If any word starts from an unquoted tilde character ('~'), then all the characters from the first encoded slash are categorised as tilde prefixes. If any of the characters in the tilde prefix are quoted, then the characters are treated as a login name which is a null string. If the prefix is '~+', then the value of its variable can replace the tilde prefix. And if the prefix is '~-' the value of the variable gets substituted, i.e. if it is set.

Any character following the tilde in Tilde prefix has a number N and it may also be prefixed with '+' or '-'. From the directory stack, this prefix gets replaced with the corresponding element and displayed by a built-in system which is invoked by the characters following this prefix as an argument. If Tilde prefix is sans tilde then it consists of a number and '+', '-' is assumed. When the login name is found invalid, then the Tilde expansion fails, and the word is left unchanged.

Arithmetic expansion – This expansion helps to utilise arithmetic and support the evaluation of any arithmetic expression alongside its substitution for the result. The format of this expansion is as follows-

- `$ ((expression))`

The expression in this expansion is treated as if it is in double quotes, but the quote inside the parentheses is not treated specifically. Tokens in this expres-